1. Main question

Does cultural background (Mexico vs. United States) influence whether individuals defer to AI or human expert recommendations when the two conflict, and is this effect driven by perceived legitimacy (authority) rather than perceived competence? Prior work suggests that deference is typically driven by perceived competence (Hou & Jung, 2021), but cultural factors such as power distance may shape how authority is evaluated (Wang, 2025).

2. Key dependent variables and how they are measured

Primary dependent variable:

- Advisor choice: single item, binary forced choice per scenario ("Whose recommendation would you follow?"; AI = 0, Human = 1).

Secondary dependent variable:

- Confidence: single-item, 7-point Likert scale (1 = not at all confident, 7 = extremely confident).

Scenario-level predictors:

- Relative competence: single bipolar item ("Compared to each other, which source is more likely to be correct in this situation?"; 1 = definitely the AI, 7 = definitely the human expert).
- Relative legitimacy: single bipolar item ("Regardless of which recommendation you personally agree with, which source has more right or authority to advise in this situation?"; 1 = definitely the AI, 7 = definitely the human expert).

Pre-scenario measures:

- AI Authority Perception Scale (3 items): 7-point agreement scale (1 = strongly disagree, 7 = strongly agree), including one reverse-scored item; final score is the mean of all items after reverse scoring.
- Authority Criteria Scale (4 items): 7-point importance ratings (1 = not at all important, 7 = extremely important). Social/role-based authority is calculated as the mean of items assessing credentials, accountability, and humanness; performance-based authority is measured using the item assessing accuracy.

Higher scores indicate greater endorsement of the construct.

3. Conditions

Between-subjects factor:

- Cultural group (measured): Mexico = 1, United States = 0.

Within-subjects factor:

- Domain: technical (medical diagnosis, financial advice) vs. interpersonal (mental health, career advice).

All participants complete four scenarios (two per domain) in counterbalanced order.

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4. Assignment to conditions

Participants are not randomly assigned to cultural group; this is a measured variable based on country of residence. All participants complete all scenario types (within-subjects design), with order counterbalanced using a Latin square design.

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5. Main analyses

All analyses will be conducted using mixed-effects models.

Variable treatment

- Advisor choice (DV) is binary (AI = 0, Human = 1).
- Cultural group is dummy coded (United States = 0, Mexico = 1).
- Domain is effect coded (technical = −0.5, interpersonal = +0.5).
- Relative legitimacy and relative competence are treated as continuous predictors (1–7 scales) and will be mean-centered within participants to separate within-person effects from between-person differences.
- Continuous covariates (religiosity, AI familiarity, and age) will be z-scored (standardized) prior to analysis.
- Education will be treated as an ordinal variable and standardized (z-scored).

Primary analysis
Advisor choice will be analyzed using a mixed-effects logistic regression model.

Fixed effects:

- Cultural group
- Domain
- Cultural group × domain interaction
- Covariates: religiosity, education, AI familiarity

Random effects:

- Random intercepts for participants
- Random slopes for domain by participant will be included; if the model fails to converge, the random slopes will be removed and only random intercepts retained

A significant positive coefficient for cultural group (Mexico > United States) will indicate greater likelihood of selecting the human expert.

Decision-rule analysis (critical test)
A second mixed-effects logistic regression will test whether reliance on legitimacy versus competence differs by cultural group.

Fixed effects:

- Relative legitimacy (centered)
- Relative competence (centered)
- Cultural group
- Cultural group × legitimacy
- Cultural group × competence

Random effects:

- Random intercepts for participants

A significant cultural group × legitimacy interaction, such that legitimacy more strongly predicts advisor choice for Mexican participants than for U.S. participants, will support the hypothesis that higher power-distance contexts rely more on authority when making decisions. A smaller or non-significant interaction for competence would be more consistent with this authority-based account than with a competence-based account.

Supplementary analysis
A linear mixed-effects model will test whether perceived legitimacy differs by cultural group and domain.

Outcome:

- Relative legitimacy

Fixed effects:

- Cultural group
- Domain
- Cultural group × domain interaction
- Covariates: religiosity, education, AI familiarity

Random effects:

- Random intercepts for participants

A significant effect of cultural group, such that Mexican participants assign greater legitimacy to the human expert relative to U.S. participants, will support the idea that cultural group influences authority judgments.

A power distance effect will be supported only if:

1. Cultural group significantly predicts advisor choice.
2. Cultural group significantly predicts perceived legitimacy.

3. Perceived legitimacy significantly predicts advisor choice.
4. Cultural differences in advisor choice are explained by legitimacy but not competence.
5. Legitimacy is a stronger predictor of choice for Mexican participants than for U.S. participants.

6. Outliers and exclusions

Exclusions will be applied to a blinded dataset prior to analysis.

Participant-level exclusions:

- Failed attention check
- Completion time < 5 minutes
- Self-reported professional AI or machine learning expertise
- Country of residence mismatch

Trial-level exclusions:

- More than 10% missing responses
- Conflict perception score < 4 (that trial excluded only)
- All primary analyses will be repeated including only trials in which participants report high conflict perception (≥ 6). This ensures that results are not driven by participants who did not perceive a meaningful disagreement between sources.

Outliers:
Age will be screened at ±3 standard deviations and retained unless clearly due to error.

7. Sample size

Target N = 240 participants to obtain approximately N = 200 after exclusions (100 per country). This is based on an a priori power analysis (odds ratio ≈ 1.5, $\alpha = .05$, power = .80). Data collection will stop once N = 120 is reached per country.

8. Additional details

Covariates include religiosity, education, and AI familiarity to control for alternative explanations. Scenario order will be counterbalanced. All measures are administered in a fixed sequence within each scenario to reduce demand effects.

Hou & Jung (2021). *Proceedings of the ACM on Human-Computer Interaction*, *5*(CSCW2), 1–25. https://doi.org/10.1145/3479864

Wang, S. (2025). Public perceptions of artificial intelligence in 20 countries: Assessing individual- and country-level factors. *Cross-Cultural Research, 59*(5), 651–676. https://doi.org/10.1177/10693971251336803